

Counterexample to Finite-Dimensional Isometric Universality of $\text{Lip}[0, 1]$

Run `fa_banach_001`, agent `agent_lane_04`

June 15, 2026

Source Question

The source is M. A. Sofi, *Embeddability, representability and universality involving Banach spaces*, arXiv:1907.07369. On page 8, after discussing Raja's theorem that closed subspaces of $C[0, 1]$ consisting entirely of Lipschitz functions are finite dimensional, the paper asks:

Problem 5. Is it true that each finite dimensional Banach space can be isometrically embedded into $\text{Lip}[0, 1]$?

We conclude with the following natural problems which to the best of our knowledge remain open.

Problem 4. Is it true that each finite dimensional Banach space can be isometrically embedded into $D[0, 1]$?

Problem 5. Is it true that each finite dimensional Banach space can be isometrically embedded into $\text{Lip}[0, 1]$?

Throughout this packet, $\text{Lip}[0, 1]$ is interpreted in the surrounding sense of the paper: Lipschitz functions regarded as a subset of $C[0, 1]$, with the inherited supremum norm. This is the interpretation suggested by the immediately preceding question about closed subspaces of $C[0, 1]$ consisting entirely of Lipschitz functions.

Claimed Counterexample

The answer to Problem 5 is negative. The three-dimensional Hilbert space ℓ_2^3 does not admit a linear isometric embedding into $\text{Lip}[0, 1]$ when $\text{Lip}[0, 1]$ carries the inherited $C[0, 1]$ -norm.

Proof Intuition

A linear subspace of $C[0, 1]$ is encoded by its evaluation functionals. If $T : \ell_2^3 \rightarrow C[0, 1]$ is a linear isometry and every Tx is Lipschitz, then those evaluation functionals move along a Lipschitz curve in the dual unit ball of ℓ_2^3 . But to recover the Euclidean norm exactly, the norming set must contain every point of the Euclidean dual sphere, up to sign. A Lipschitz curve cannot cover a two-dimensional sphere.

Theorem and Proof

Theorem 1. *There is no linear isometry*

$$T : \ell_2^3 \longrightarrow C[0, 1]$$

such that Tx is Lipschitz on $[0, 1]$ for every $x \in \ell_2^3$.

Proof. Suppose that such an isometry T exists. Let e_1, e_2, e_3 be the standard orthonormal basis of ℓ_2^3 , and define

$$F(t) = ((Te_1)(t), (Te_2)(t), (Te_3)(t)) \in \mathbb{R}^3, \quad 0 \leq t \leq 1.$$

Since each Te_j is Lipschitz, the map $F : [0, 1] \rightarrow \mathbb{R}^3$ is Lipschitz.

For every $x = (x_1, x_2, x_3) \in \mathbb{R}^3$,

$$(Tx)(t) = \sum_{j=1}^3 x_j (Te_j)(t) = \langle x, F(t) \rangle.$$

Because T is an isometry into $C[0, 1]$,

$$\|x\|_2 = \|Tx\|_\infty = \sup_{0 \leq t \leq 1} |\langle x, F(t) \rangle|.$$

First, this implies $\|F(t)\|_2 \leq 1$ for every t , since otherwise the functional $x \mapsto \langle x, F(t) \rangle$ would have norm greater than 1.

Now fix any $u \in S^2$, the unit sphere of $\mathbb{R}^3$, and apply the displayed identity to $x = u$. Compactness gives a point $t \in [0, 1]$ such that

$$|\langle u, F(t) \rangle| = 1.$$

Since $\|u\|_2 = 1$ and $\|F(t)\|_2 \leq 1$, equality in Cauchy's inequality forces $F(t) = u$ or $F(t) = -u$. Hence

$$S^2 \subset F([0, 1]) \cup (-F([0, 1])).$$

This is impossible. A Lipschitz image of an interval has Hausdorff dimension at most 1, so the right-hand side has Hausdorff dimension at most 1. But the Euclidean sphere S^2 has Hausdorff dimension 2. This contradiction proves the theorem. $\square$

Consequences

Under the inherited-supremum-norm reading of $\text{Lip}[0, 1]$, the three-dimensional Hilbert space is a finite-dimensional Banach space that cannot be isometrically embedded into $\text{Lip}[0, 1]$. Therefore Problem 5 has a negative answer in that sense.

Limitations

This packet does not address Problem 4 about the space $D[0, 1]$ of everywhere differentiable functions. The same one-dimensional-image obstruction is immediate for C^1 parametrizations, but the source writes $D[0, 1]$, not $C^1[0, 1]$, and everywhere differentiable curves can be subtler than Lipschitz curves. The packet also does not address conventions under which $\text{Lip}[0, 1]$ is equipped with a Lipschitz norm rather than the inherited $C[0, 1]$ -norm.

Novelty and Duplicate Checks

Before promotion I searched the cheap run indexes for arXiv 1907.07369, the title terms, and core phrases including $\text{Lip}[0,1]$, finite-dimensional Banach spaces, isometric embeddings, Lipschitz functions, embeddability, representability, and universality. No exact packet or attempt hit was present; broad universality and embedding hits were unrelated.

I also ran bounded web searches for exact and near-exact phrases such as

- "Problem 5" "Lip[0,1]" "finite dimensional Banach space",
- "each finite dimensional Banach space" "Lip[0,1]",
- "isometrically embedded into Lip[0,1]",
- finite-dimensional Banach spaces, Lipschitz functions, isometric embeddings, and $C[0,1]$.

No later paper explicitly answering Sofi's Problem 5 was found in this bounded search. The argument above is elementary and should still be reviewed for the intended norm convention.

References

References

- [1] M. A. Sofi, *Embeddability, representability and universality involving Banach spaces*, arXiv:1907.07369.
- [2] J. Heinonen, *Lectures on Analysis on Metric Spaces*, Universitext, Springer, 2001. Used for the standard facts that Lipschitz maps do not increase Hausdorff dimension and that the Euclidean sphere S^2 has Hausdorff dimension 2.

Human Review Recommendation

This is a short candidate counterexample. Review should focus on two points: whether the source paper indeed intends the inherited $C[0,1]$ -norm on $\text{Lip}[0,1]$, and whether the evaluation-functional norming argument is accepted as forcing the whole Euclidean dual sphere to lie in the union of the two Lipschitz curve images.